

In-Context Learning Below One Percent: Tabular Foundation Models versus Gradient-Boosted Trees for Card Fraud Detection

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SMAI Course Project Proposal, Monsoon 2026

1 Introduction

Gradient-boosted decision trees (GBDTs) – **XGBoost** and its widely used derivatives **LightGBM** and **CatBoost** – are the default, **traditional/classical ML** choice for tabular prediction. Every new dataset requires a preprocessing pipeline, a hyperparameter search and a training run. Tabular foundation models (TFMs) such as TabPFN v2 [1] and TabICLv2 [2] remove that loop. They are pretrained once on millions of synthetic tabular tasks and then treat a labelled table as a prompt: a context set $\{(x_1, y_1), \dots, (x_k, y_k)\}$ is passed through the network alongside a query x_* , and the prediction comes out of a single forward pass. This mechanism, **in-context learning (ICL)**, is the tabular analogue of prompting an LLM with examples rather than fine-tuning it: the rows in the context play the role the exemplars play in a text prompt, and which rows go in is the only lever left once the weights are frozen. Mechanically, TFMs are instances of **Prior-Data Fitted Networks (PFNs)**: transformers trained offline to approximate **Bayesian** posterior-predictive inference under a synthetic prior over data-generating processes, so that a single forward pass at test time is standing in for an otherwise intractable Bayesian update. The weights are never updated.

We use **TabICLv2** rather than the original TabICL: it is **open-source** (weights and inference code public, with the full pretraining pipeline to follow), reports **SOTA** results among current TFMs on TabArena and TALENT, and natively handles a **context window of up to 60K rows** during pretraining [2], comfortably covering the $k \in \{1, 2, 5, 10\}K$ context budgets swept in this project. Both TabPFN v2 and TabICLv2 are **pip-installable PyPI packages** (`pip install tabpfn`, `pip install tabicl`) and both expose a **fine-tuning** interface, useful for exploration beyond zero-shot ICL if time permits.

Why attention, not just trees. Attention-based TFMs are worth testing against **XGBoost**-style GBDTs not because trees are weak – they are the strongest **classical ML baseline** on tabular data – but because the two families adapt differently: a GBDT commits to fixed axis-aligned splits at training time and needs a retrain for a new pattern, while an attention-based TFM recomputes feature interactions and row relevance on the fly, so adaptation means changing the context, not re-fitting parameters. In fraud, where the signal is a shifting combination of features, that difference is the whole reason ICL is worth testing here rather than a curiosity next to a GBDT that already works well.

Published results show TFMs matching or beating tuned GBDTs on standard benchmarks [1]. This project asks whether the same holds for card fraud detection. Section 2 argues that the way those benchmarks are built leaves the question open.

Goal. Determine, under a matched-budget protocol, whether ICL keeps its advantage over gradient boosting when the positive rate falls below 1%, and identify which rows should be placed in the context when positives are rare – in short, whether *context engineering* for anomaly detection is a real design problem or a marginal one.

2 Research Gap

The benchmarks used to establish the TFM result exclude the fraud regime by construction, and the benchmark that covers that regime does not include TFMs.

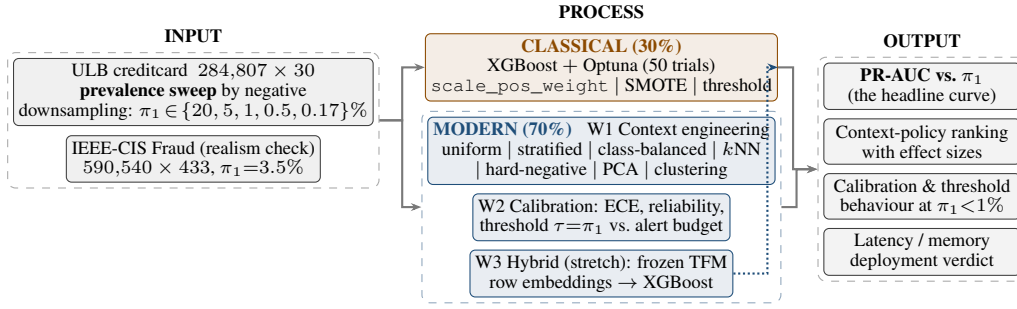
Benchmark	Headline metric	Minority-rate floor	TFMs incl.?
OpenML-CC18	not specified	$\geq 5\%$ (hard curation rule)	yes
TALENT	Accuracy, F1, AUC	imbalanced sets excluded	yes
TabArena	ROC-AUC	no imbalance criterion stated	yes
CLIMB [4]	AUPRC	IR to 577.9 (0.17%)	no
This project	PR-AUC	0.17%	yes

Three points follow. First, published claims that TFMs beat GBDTs rest on accuracy or ROC-AUC, and ROC-AUC is prevalence-invariant, so it stays high even when false positives greatly outnumber true positives. Second, the only systematic study of imbalance in prior-data-fitted networks [5] stops at a 5% minority rate, roughly 29 times more prevalent than card fraud, and does not report PR-AUC. Third, work on tabular ICL example selection does exist, covering retrieval-based context selection and comparisons of uniform, K -means and farthest-point sampling, but neither line of work considers class imbalance. Class-aware context construction therefore remains unexamined.

One paper should be addressed directly. García et al. [3] evaluate TabPFN and TabICL on intrusion detection with classes as rare as 0.0004% and report that TFMs recover minority classes on which tree ensembles fail. Their protocol supplies the TFMs with class-balanced contexts while training the GBDT baselines on raw imbalanced data, so model family is confounded with sampling strategy, and they report per-class recall rather than PR-AUC. Their finding is a reason to run this study rather than an answer to it, and separating the confound is one contribution of the project.

What is different here. No prior work evaluates ICL against tuned GBDTs on PR-AUC below 1% prevalence with the sampling budget held equal across model families, and no prior work treats *how a context is built* as the primary variable rather than an implementation detail. That is the gap this project fills: fraud is the regime where existing TFM benchmarks stop and existing imbalance benchmarks skip the model family that matters most.

Research questions. **RQ1:** Does the ICL advantage survive as π_1 falls from 20% to 0.17%, measured in PR-AUC? **RQ2:** Does the context-construction policy matter more than the choice of model family? **RQ3:** Does the correction $\tau = \pi_1$, validated only to 5% [5], extend two orders of magnitude further, and does TabPFN’s calibration hold there?



TabPFN v2 / TabICLv2: inference only, ensembled over M context subsets

Figure 1: Both arms share identical splits and a single scoring harness; the **classical (traditional ML)** arm gets the same budget k as the modern arm, removing the confound in [3]. W4 (Section 4.1) sweeps dataset scale and shot count across both arms and is described in text rather than boxed here.

3 Overview

4 Approach

4.1 Modern component (70% of effort)

TabPFN v2 and TabICLv2 are used with frozen weights, so the work lies in how they are supplied with data. TabPFN v2 accepts roughly 10K context rows, 500 features and 10 classes [1], and TabICLv2 scales further, to a **60K-row** context window [2]. Neither can hold 284K transactions, so the context has to be chosen, and at $\pi_1 = 0.172\%$ a uniformly drawn 10K-row context contains about 17 positives. Since ICL does no gradient-based adaptation, every improvement has to come from *context engineering*: choosing which rows enter the prompt, in what proportion, and how many prompts to ensemble – the tabular counterpart of prompt engineering for LLMs, but with labelled transactions in place of exemplar text.

- **W1. Class-aware context construction (core contribution).** Compare uniform, prevalence-preserving stratified, class-balanced, k NN-retrieved (query-conditional), hard-negative-mined, **PCA**-projected (span the leading principal components of the minority class) and **clustering**-based (per-class k -means/GMM centroids and boundary exemplars) policies, extending the uniform/ K -means comparisons in prior work to the class-imbalanced setting they skip. Sweep the budget $k \in \{1, 2, 5, 10\}$ K and cover the full dataset by ensembling M disjoint contexts, aggregating in log-odds space. Earlier work reports that more elaborate selection barely improves on uniform sampling for balanced data; whether that holds when positives are scarce – where a uniform context may contain only a handful of frauds – has not been tested.
- **W2. Calibration and decision rule.** PFNs are reported to be well calibrated, which is the stated reason that thresholding at $\tau = \pi_1$ is sufficient at moderate imbalance [5]. We test this at $\pi_1 = 0.0017$ using reliability diagrams and expected calibration error, and compare $\tau = \pi_1$ with F1-optimal and alert-budget thresholds.
- **W3. Hybrid representation transfer (stretch goal).** TabPFN v2 produces a row-embedding space that is close to linearly separable. Feeding frozen embeddings to XGBoost separates the contribution of the learned representation from that of the in-context inference procedure. This workstream is scheduled last and may be dropped (Section 9).
- **W4. Scale and shot-count study (extended).** Two axes complete the picture beyond prevalence. *Dataset scale*: holding π_1 fixed, grow the training pool N toward the full 284K rows and track whether the TFM-GBDT gap widens, narrows or reverses, since GBDTs keep improving with more rows while a TFM’s context is capped. *Shot count*: the budget k swept in W1 doubles as a few-shot-to-many-shot axis – $k \approx 100$ –1K is **few-shot**, $k = 10$ K is **many-shot** – reporting where PR-AUC saturates, in place of a zero-shot framing that does not apply once a labelled context is required. Both axes pair with the **training time** and **inference time** comparison in Section 8, feeding the deployment verdict in Section 7.

4.2 Classical component (30% of effort)

A tuned XGBoost baseline, set up to be as strong as possible so that any TFM advantage is credible. The pipeline uses `scikit-learn` with target encoding and median or indicator imputation. Imbalance is handled three ways: `scale_pos_weight`, SMOTE, and plain fitting with post-hoc threshold tuning. A 50-trial Optuna TPE search covers depth, learning rate, subsampling and γ, α, λ , early-stopped on validation average precision. Matched-budget arms, in which XGBoost is trained on the same k -row subsets supplied to the TFM, separate the effect of model family from the effect of sample size.

5 Datasets and Protocol

The primary benchmark is **ULB Credit Card Fraud** (Kaggle `mlg-ulb`; 284,807 transactions, 492 frauds, $\pi_1 = 0.172\%$, 30 PCA-anonymised numeric features). The prevalence axis is produced inside this dataset by downsampling negatives to $\pi_1 \in \{20, 5, 1, 0.5, 0.172\}\%$, which holds features, positives and preprocessing fixed and leaves π_1 as the only quantity that varies. This costs less than assembling several datasets and avoids confounding prevalence with dataset identity. It also continues the axis on which [5] stops at 5%. **IEEE-CIS Fraud Detection** (590,540 transactions, $\pi_1 \approx 3.5\%$, 433 features after joining the identity table) serves as a single realism check, supplying high-cardinality categoricals, heavy missingness and a feature count near TabPFN’s 500 ceiling. Both datasets are public and already downloaded. Splits are stratified and fixed once, shared byte-for-byte across arms, with the test set held back until the end.

6 Evaluation

Accuracy carries no information here, since predicting no fraud on every transaction scores 99.83%, and ROC-AUC alone is optimistic under imbalance, which is why it is reported alongside PR-AUC below rather than in place of it. The following **fraud metrics** anchor every comparison:

- **Primary: PR-AUC** (average precision) as the headline metric, reported alongside **ROC-AUC** for comparability with published

TFM benchmarks despite its known prevalence-invariance, and **recall at a top-1% alert threshold** (i.e., recall when flagging only the highest-risk 1% of transactions) as the metric closest to how an alert queue is actually sized.

- **Operational:** precision@ k at a 0.1% alert budget, recall at $\text{FPR} = 0.1\%$, and F1 at the tuned threshold. The alert budget follows from practice, since investigators can typically verify only on the order of 100 cards per day.
- **Calibration:** expected calibration error and reliability diagrams, which are central to RQ3.
- **Efficiency:** per-record inference latency and peak GPU memory, compared against tree traversal, measured across the W4 dataset-scale sweep so the comparison is a curve against N rather than a single number.
- **Statistical treatment:** five stratified repeats reported as mean $\pm$ std, with paired bootstrap tests. The dataset contains 492 positives, so single-seed differences cannot be interpreted.

Success criteria. The project succeeds if it produces (i) a reproducible matched-budget comparison across the prevalence sweep, (ii) a curve of PR-AUC against π_1 showing where the two families change order, and (iii) a ranked comparison of context-construction policies with effect sizes. A negative result with a supported explanation, for example that the ICL advantage disappears below 1% together with the mechanism behind it, counts as a success. An inconclusive or unfair comparison does not.

7 Expected Impact

Scientific. This would provide the first PR-AUC evaluation of tabular foundation models against tuned GBDTs below 1% prevalence. It extends the imbalance literature roughly 29 times below its current floor [5] and fills the empty cell in Table 1 by measuring TFMs on the axis CLIMB uses [4]. The context-engineering study addresses a question the tabular ICL literature has not yet posed: for anomaly detection specifically, is the context-construction policy a bigger lever than model choice, and the matched-budget protocol tests whether the strongest current pro-TFM claim [3] holds under a fair comparison.

Practical. Industry estimates put worldwide card fraud losses in the tens of billions of dollars annually, with continued growth projected. The limiting resource in deployment is labelled data, not compute: chargeback confirmation arrives days to weeks after a transaction, and only a small fraction is ever verified. A model needing no training and only a few thousand labelled examples as context speaks directly to that constraint, and to cold-start settings such as a new market with little labelled fraud. Good context engineering is then effectively free feature engineering, substituting for the retraining cycle a GBDT needs whenever the fraud pattern drifts. If accuracy holds but latency does not, the result is still useful: tabular ICL has been reported to run orders of magnitude slower than LightGBM for single-record scoring against millisecond production budgets. The project ends with a measured verdict on where in a fraud stack a TFM can be placed, inline at authorisation or in the offline review queue.

8 Compute

Primary compute is **IIT-H’s ADA cluster**, with **4 GPUs in parallel** – one per context policy or budget in W1/W4 – so the sweep completes in a single batch instead of sequentially, leaving room to iterate toward the strongest possible XGBoost baseline and to compare **training time** and **inference time** trade-offs for both model families directly. **Google Colab** (single T4, 16 GB) is the fallback, since TFM cost is bounded by context size, not dataset size: a 10K-row context fits under 4 GB, so the full ULB sweep is feasible there too. Total estimated cost is under 30 GPU-hours.

9 Execution Plan and Scope Control

This is a single-person project, so the scope is tiered and the tiers are respected. Core work, which must be completed, covers the data pipeline, the tuned XGBoost baseline, TFM inference, the prevalence sweep and W1. Extended work covers W2, W4, the IEEE-CIS realism check and latency profiling. Stretch work covers W3, hard-negative mining, and extending W4’s scale sweep to IEEE-CIS. If time runs short the stretch items are dropped without affecting the core claim, and the prevalence sweep degrades by removing its largest- N points, which shortens the curve without invalidating it. The schedule is feasible because the modern arm needs no training, which removes the most expensive part of a conventional deep learning project.

10 Timeline

Weeks	Milestone	Deliverable
1–2	Both arms live	Splits fixed. XGBoost baseline and TabPFN v2 inference both producing PR-AUC on ULB. The TFM needs no training, so it runs in week 1.
3–4	Prevalence sweep	Controlled- π_1 datasets built, matched-budget arms in place, first curve of PR-AUC against π_1 .
5–6	Mid-evaluation (10 Oct)	Core result in hand: both families across the full prevalence sweep, plus an initial W1 policy comparison.
7–9	W1, W2, W4	Full context-policy study with repeats, calibration and threshold analysis, dataset-scale and shot-count sweep, IEEE-CIS run.
10–11	W3 and analysis	Hybrid embeddings if on schedule, latency profiling, statistical testing.
12	Write-up	Final report, cleaned repository, presentation video.

The **classical (traditional ML) baseline** is usually built first, but doing so would leave 70% of the project unstarted at the mid-evaluation. Since the TFM needs no training, both arms can run from week 1 and the mid-evaluation can present the comparison itself.

References

- [1] N. Hollmann et al. Accurate predictions on small data with a tabular foundation model. *Nature* 637:319–326, 2025.
- [2] J. Qu, D. Holzmüller, G. Varoquaux, M. Le Morvan. TabICLv2: A better, faster, scalable, and open tabular foundation model. arXiv:2602.11139, 2026.
- [3] J. García, J. de Curtò et al. Foundation models for cybersecurity: A multi-modal evaluation of TabPFN and TabICL for tabular intrusion detection. *Electronics* 14(19):3792, 2025.
- [4] CLIMB: A benchmark of class-imbalanced learning methods on tabular data. arXiv:2505.17451, 2025.
- [5] A. McDowell, A. Stromberg, R. Sankar. Correcting class imbalance in prior-data fitted networks for tabular classification. arXiv:2605.21742, 2026.